Electronic Circuit Design

Lesson 1: Operational Amplifier Fundamentals

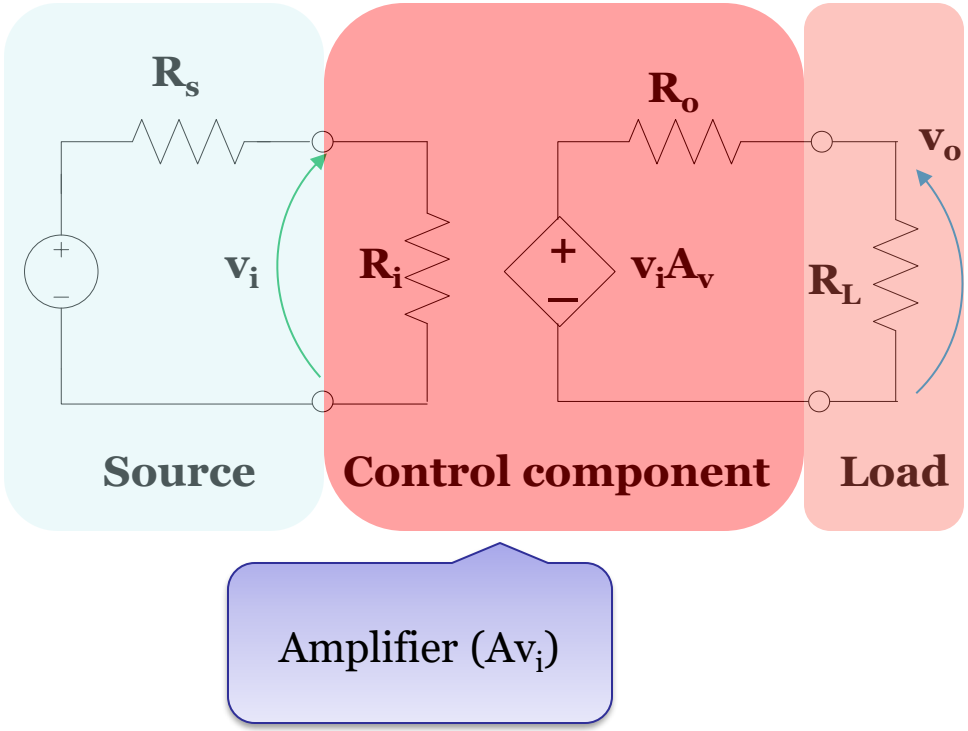
Key Terms

- voltage amplifier
- inverting amplifier
- non-inverting amplifier
- voltage follower
- open-loop
- closed loop

Amplifier Fundamentals

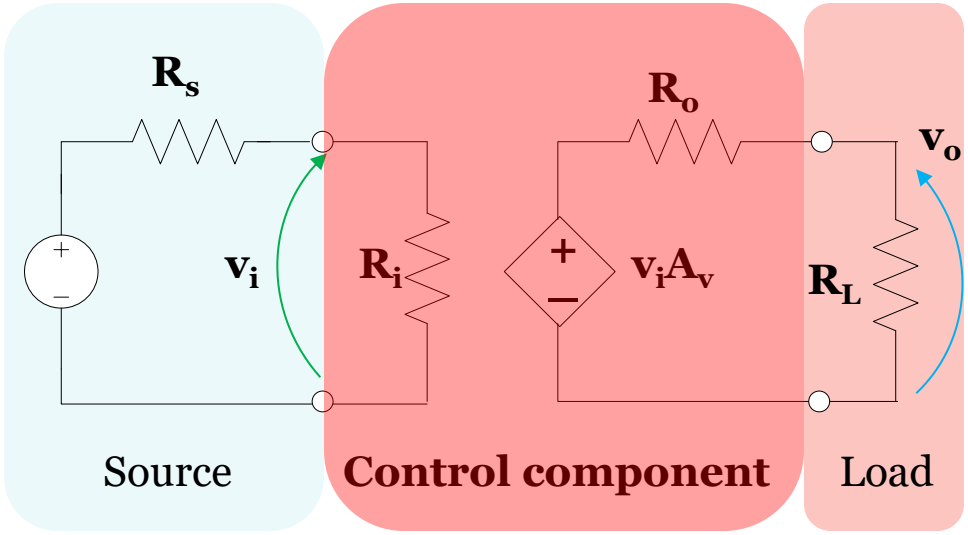
Amplifier Types/Voltage Amplifier

Voltage Amplifier: to amplify an input voltage signal and provide an output voltage signal



Amplifier Types/Voltage Amplifier

Voltage Amplifier: to amplify an input voltage signal and provide an output voltage signal



$$\frac{v_o}{v_s} = A_v \frac{R_i}{R_s + R_i} \frac{R_L}{R_o + R_L}$$

Optima the setting: Maximize $\frac{R_i}{R_s + R_i} \frac{R_L}{R_o + R_L}$

$\Rightarrow R_i >> R_s$ **and** $R_o << R_L$

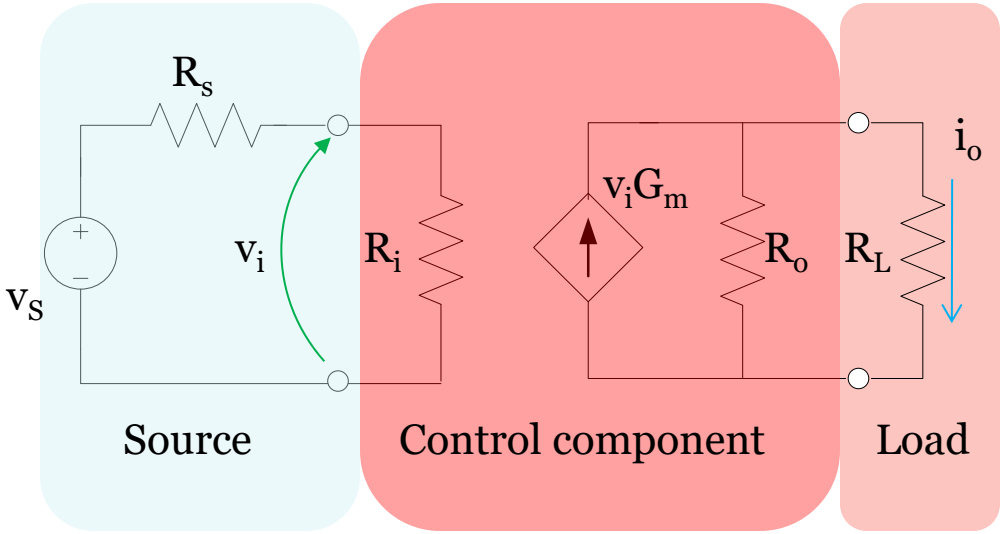
$$\Rightarrow \frac{v_o}{v_s} \approx A_v$$

$P_i = v_i i_i$ → $P_o = v_o i_o$

For an ideal voltage amplifier: $R_i \rightarrow \infty$, $R_o = 0$

Amplifier Types/Transconductance Amplifier

Transconductance Amplifier: to amplify an input voltage signal and provide an output current signal

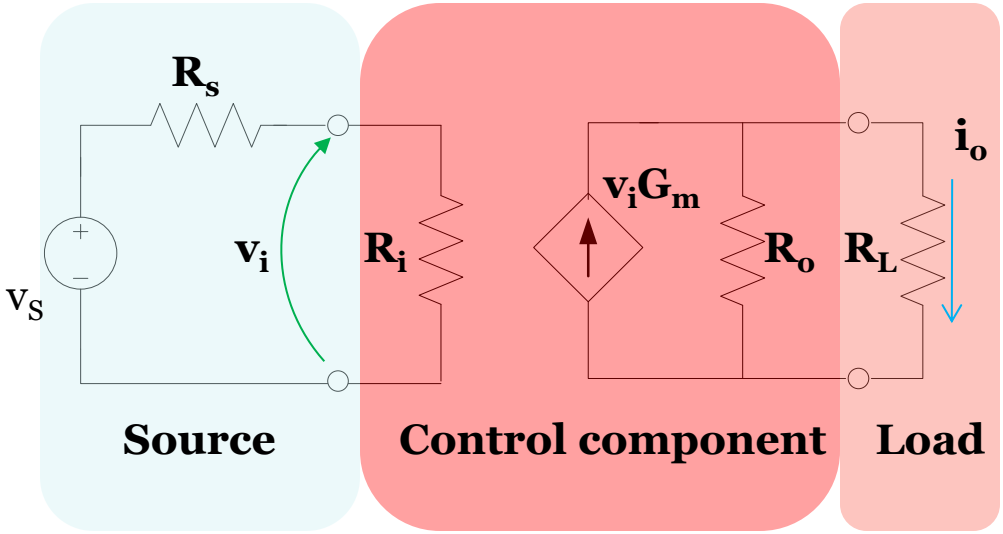


Amplifier ($G_m v_i$)

$$G_m = \frac{A_v}{R_o} \quad \text{since} \quad A_v = G_m R_o$$

Amplifier Types/Transconductance Amplifier

Transconductance Amplifier: to amplify an input voltage signal and provide an output current signal



For an ideal transconductance amplifier:
$R_i \rightarrow \infty$, $R_o \rightarrow \infty$

$$i_o = G_m v_i \frac{R_o}{R_o + R_L}$$

and

$$v_i = v_s \frac{R_i}{R_s + R_i}$$

Optima the setting:

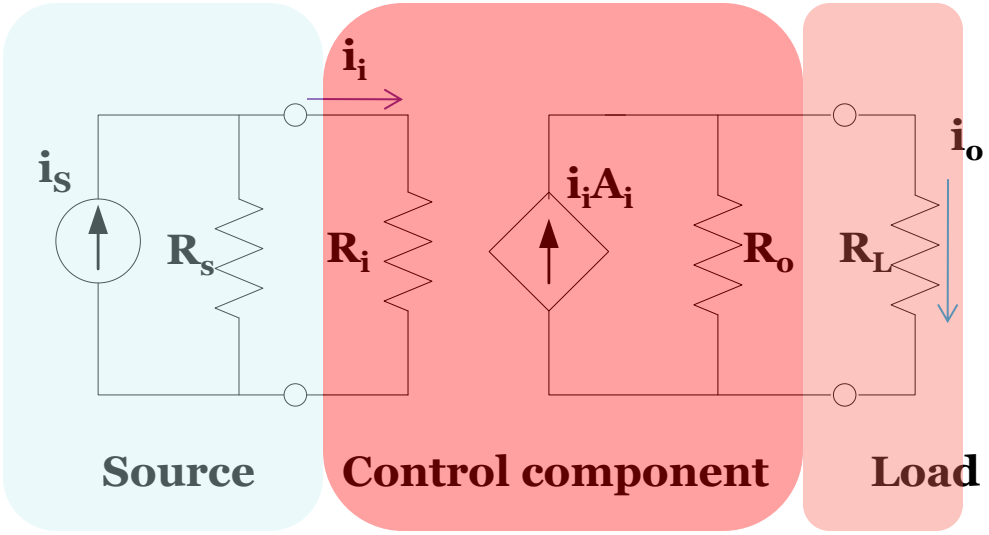
Maximize $\dfrac{R_i}{R_s + R_i} \dfrac{R_o}{R_o + R_L}$

$\Rightarrow R_i >> R_s$ and $R_o >> R_L$

$$\Rightarrow \frac{i_o}{v_i} = G_m$$

Amplifier Types/Current Amplifier

Current Amplifier: to amplify an input current signal and provide an output current signal

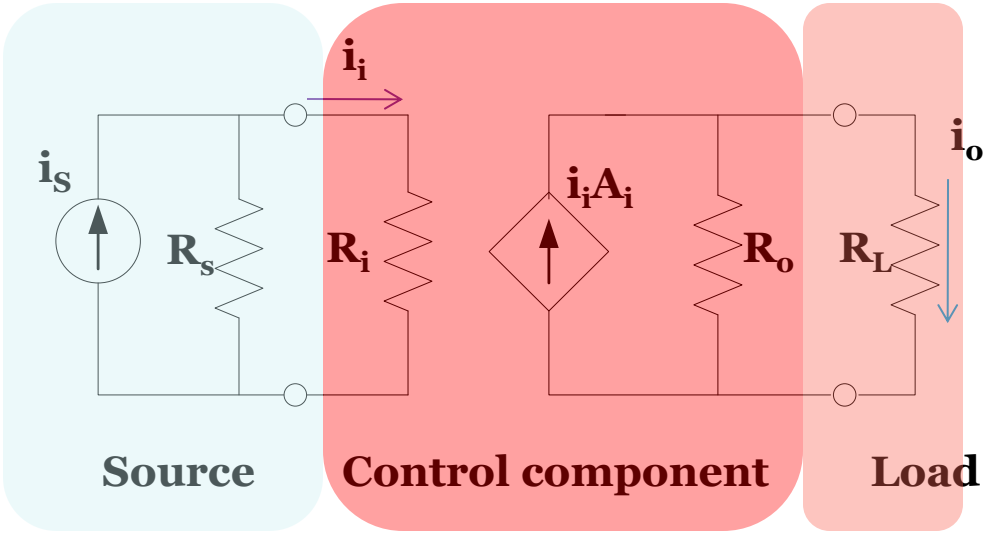


Amplifier ($A_i i_i$)

$$A_i = G_m R_i$$

Amplifier Types/Current Amplifier

Current Amplifier: to amplify an input current signal and provide an output current signal



$$i_o = A_i i_i \times \frac{R_o}{R_o + R_L}$$

and

$$i_i = i_S \times \frac{R_s}{R_i + R_s}$$

$$\Rightarrow i_o = A_i i_S \times \frac{R_o}{R_o + R_L} \frac{R_s}{R_i + R_s}$$

Current amplifiers are optimized by

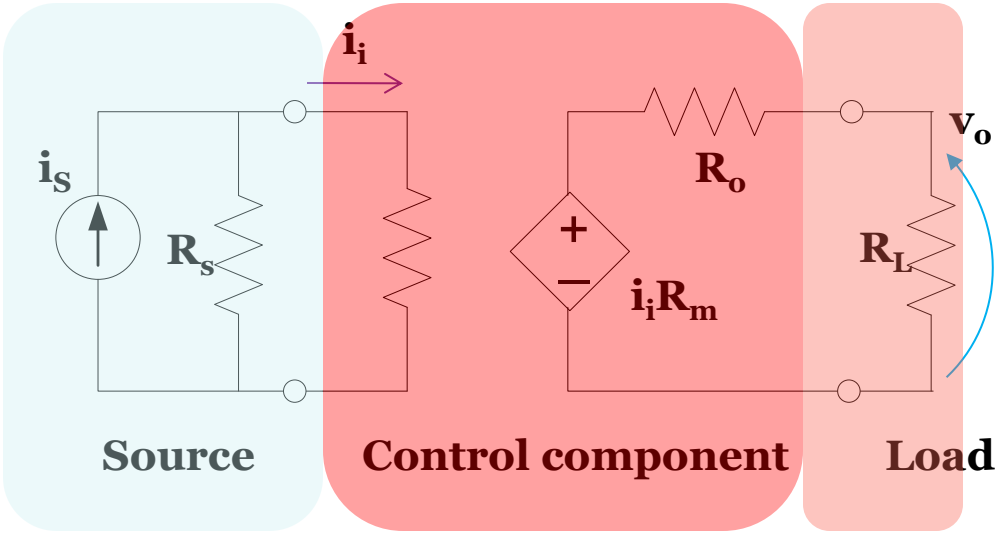
$R_i << R_s$ **and** $R_o >> R_L$

$$\Rightarrow \frac{i_o}{i_s} = A_i$$

For an ideal current amplifier: R_i=0, $R_o \rightarrow \infty$

Amplifier Types/Transresistance Amplifier

Transresistance Amplifier: to amplify an input current signal and provide an output voltage signal

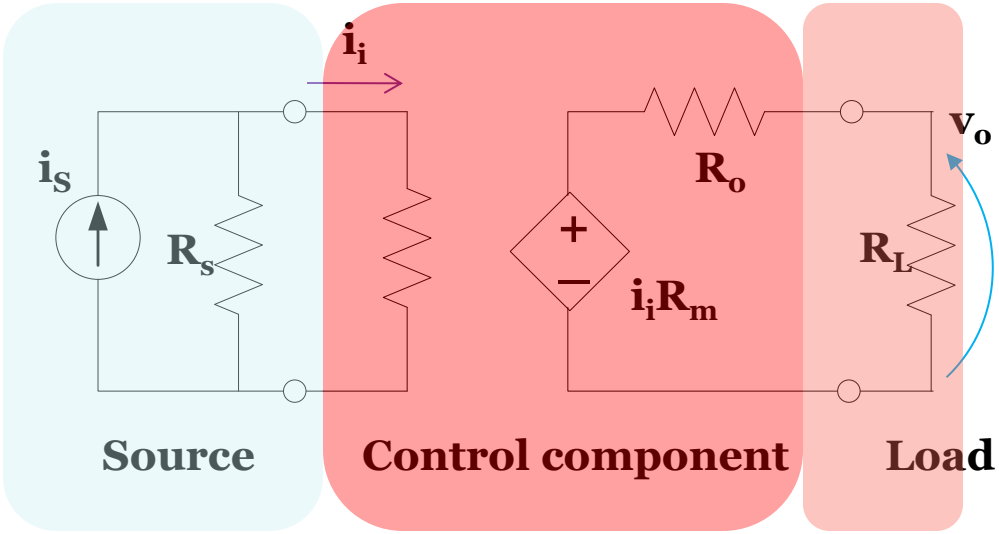


Amplifier ($R_m i_i$)

$$R_m = A_i R_o = A_v R_i$$

Amplifier Types/Transresistance Amplifier

Transresistance Amplifier: to amplify an input current signal and provide an output voltage signal



$$v_o = R_m i_i \times \frac{R_L}{R_o + R_L}$$

and

$$i_i = i_s \times \frac{R_s}{R_i + R_s}$$

Current amplifiers are optimized by $R_i << R_s$ and $R_o << R_L$

$$\Rightarrow \frac{v_o}{i_s} = R_m$$

For an ideal voltage amplifier: R_i=0, R_o=0

Amplifier Types

Example: An amplifier with R_i = 100 kΩ, A_{oc} = 100 V/V, and R_o = 1 Ω is driven by a source with R_s = 25 kΩ and drives a load
R_L = 3 Ω.

Calculate the overall gain as well as the amount of input and output loading.

Amplifier Types

Example: An amplifier with R_i = 100 kΩ, A_{oc} = 100 V/V, and R_o = 1 Ω is driven by a source with R_s = 25 kΩ and drives a load R_L = 3 Ω.

Solution

Calculate the overall gain as well as the amount of input (Ri/Ri+Rs) and output (Rload/Rload+Rout) loadings.

From: $\text{gain} = \frac{v_o}{v_s} = \frac{Ri}{R_s + R_i} AOC \frac{R_L}{R_L + R_o}$

$$\text{the overall gain} = \frac{v_o}{v_s} = \frac{100}{100+25} \times \frac{3}{1+3} = 60 \, V/V$$

$\Rightarrow$ gain reduces because of loading.

Amplifier Types

Example: An amplifier with R_i = 100 kΩ, A_{oc} = 100 V/V, and R_o = 1 Ω is driven by a source with R_s = 25 kΩ and drives a load R_L = 3 Ω.

Solution

Calculate the overall gain as well as the amount of input and output loading.

$$\text{Input loading} = \frac{100}{100+25} = 80\%.$$

$$\text{Output loading} = \frac{3}{1+3} = 75\%.$$

Operational Amplifiers

Operational amplifiers (1)

Can be configured for a variety of mathematical operations, such as addition, subtraction, differentiation, and integration.

The advent of the bipolar junction transistor led to dramatic size reduction of the op amp.

μA741 is the most popular op amp and industry standard.
Op amp hah been widely used virtually every area of analog and mixed analog-digital electronics.

Operational amplifiers (2)

The op amp is a voltage amplifier with extremely high gain.

- 741 op amp has a typical gain of 200,000 V/V
- OP-77 has a gain of 12 million V/V

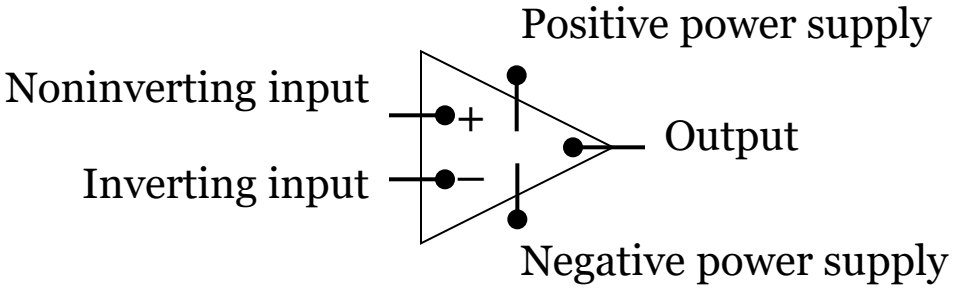
Op amp would ideally have an infinitely large gain.

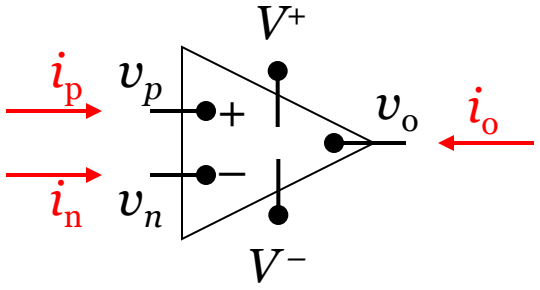
we'll start by assuming it is ideal...

Operational Amplifier (3)

Circuit Symbol and Power Supply Connections:

The op amp has a special circuit symbol, a triangle with two inputs and one output.

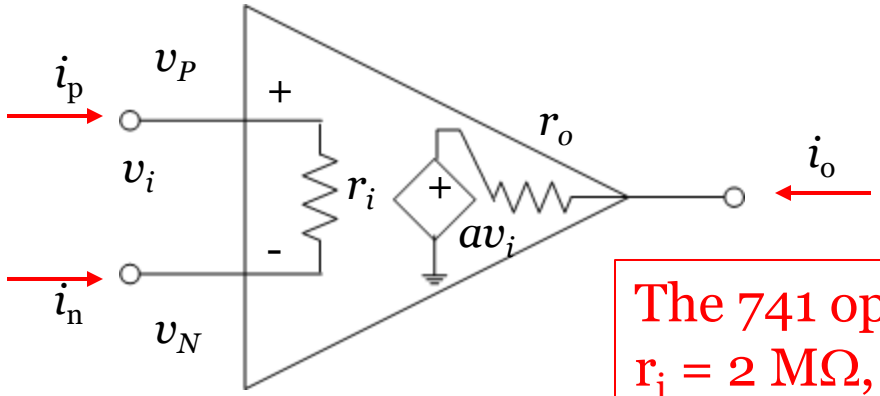




Operational Amplifier (4)

Powered Op Amp:

The 741 op amp typically has r_i = 2 MΩ, a = 200 V/mV, and r_o = 75 Ω



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Open loop parameters:

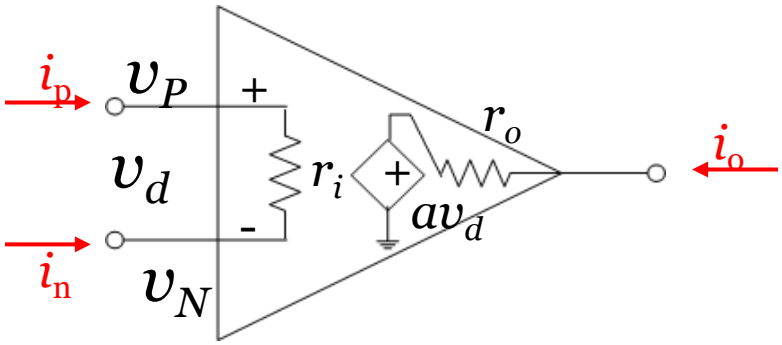
- Differential input resistance: r_i
- Voltage gain: a
- Output resistance: r_o

Operational Amplifier (5)

Powered Op Amp:

Differential input voltage:

$$v_d = v_P - v_N$$



Unloaded gain:

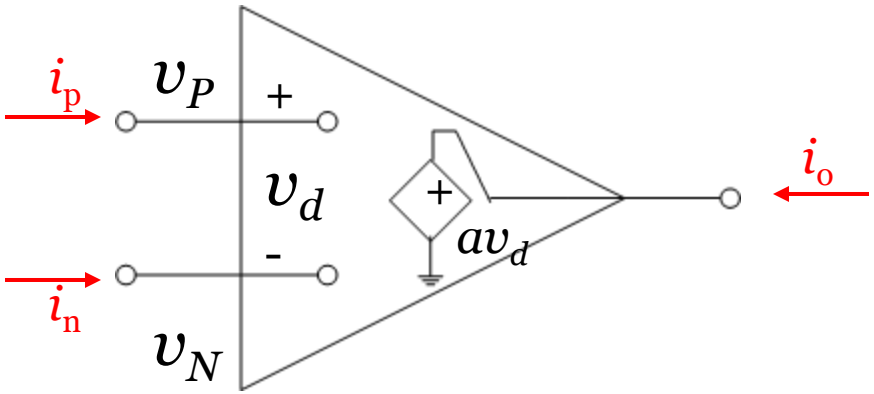
$$v_O = av_d = a(v_P - v_N)$$

$$a = \frac{v_O}{v_d}$$

Normally, v_d is a very small value.

Operational Amplifier (6)

Ideal Op Amp:



An ideal voltage amplifier with infinite open-loop gain:

$$a \to \infty$$

The terminal conditions:

$$r_i = \infty;\ r_o = 0;\ i_P = i_N = 0$$

Operational Amplifier (7)

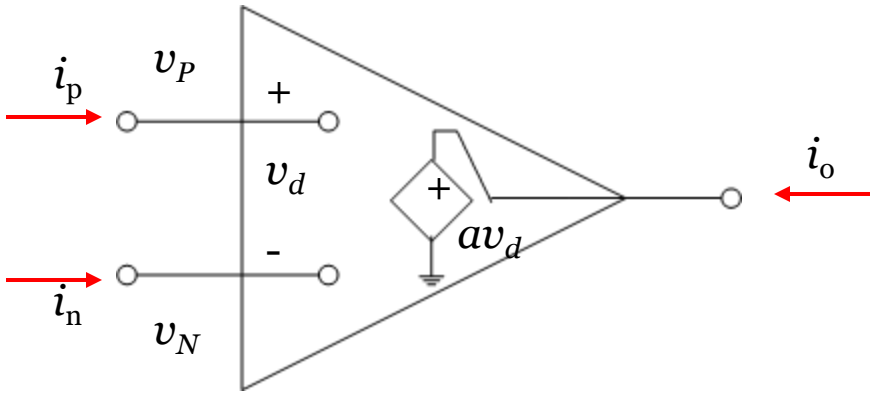
Ideal Op Amp:

Notice what this implies: If v_o is finite, then

$$v_d = \frac{v_O}{a} = \frac{v_O}{\infty} = v_P - v_N = 0$$

$$\Rightarrow v_P = v_N$$

This is called the ***virtual short*** condition.



Op Amp Configurations

Op Amp Configurations

By connecting external components around an op amp, we are constructing an op amp circuit.

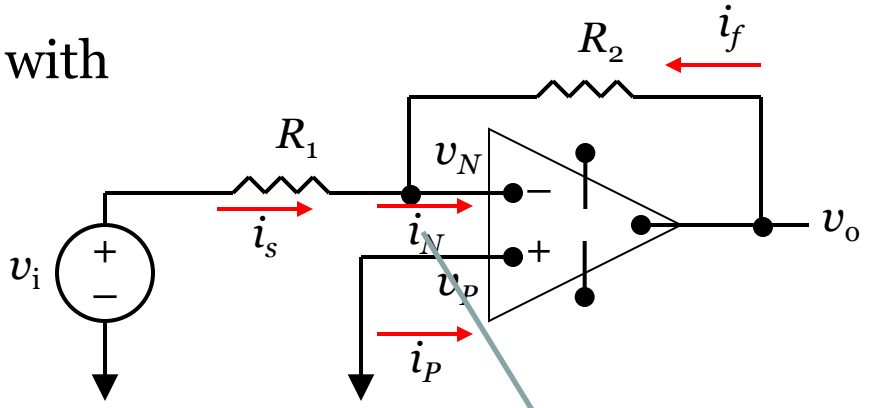
The most basic op amp circuits are ***inverting***, ***non-inverting***, and buffer.

Op Amp Configurations/Inverting Amplifier

Analyze using an op amp model with

$r_i = \infty;\ r_o = 0;$ and

gain a = *finite*



This yields

$$v_O = a(v_P - v_N) = -av_N;\ since\ v_P = 0$$

Apply KCL on negative input:

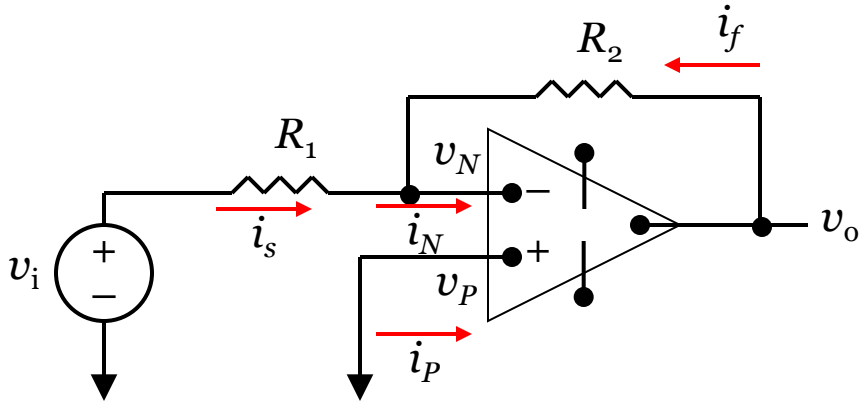
$$\frac{v_N - v_i}{R_1} + \frac{v_N - v_o}{R_2} = 0 \quad \Rightarrow \frac{R_2(v_N - v_i)}{R_1 R_2} + \frac{R_1(v_N - v_o)}{R_1 R_2} = 0$$

Op Amp Configurations/Inverting Amplifier

This yields

$$(R_1 + R_2)v_N = R_2 v_i + R_1 v_o$$

$$\Rightarrow v_N = v_i \frac{R_2}{R_1 + R_2} + v_O \frac{R_1}{R_1 + R_2}$$



From: (previous slide)

$$v_O = a(v_P - v_N) = -av_N;\ since\ v_P = 0$$

Therefore,

$$v_O = -a(v_i \frac{R_2}{R_1 + R_2} + v_O \frac{R_1}{R_1 + R_2})$$

Op Amp Configurations/Inverting Amplifier

Rearrange

$$v_o + \frac{a v_o R_1}{R_1 + R_2} = v_o\left(1 + \frac{aR_1}{R_1 + R_2}\right) = -v_i \frac{aR_2}{R_1 + R_2}$$

$$\Rightarrow \textbf{close} - \textbf{loop gain } A = \frac{v_o}{v_i} = -\frac{\frac{aR_2}{R_1 + R_2}}{1 + \frac{aR_1}{R_1 + R_2}}$$

Letting $\beta = \frac{R_1}{R_1 + R_2}$

Then

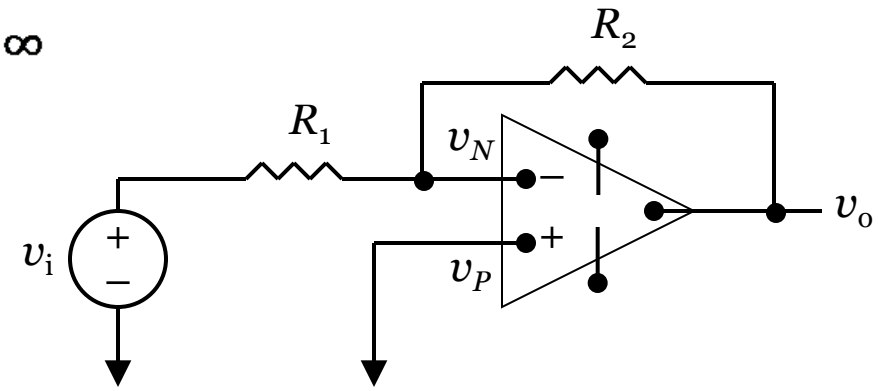
$$A = \frac{v_o}{v_i} = -a\frac{\frac{R_2}{R_1}\beta}{1 + a\beta}$$

Op Amp Configurations/Inverting Amplifier

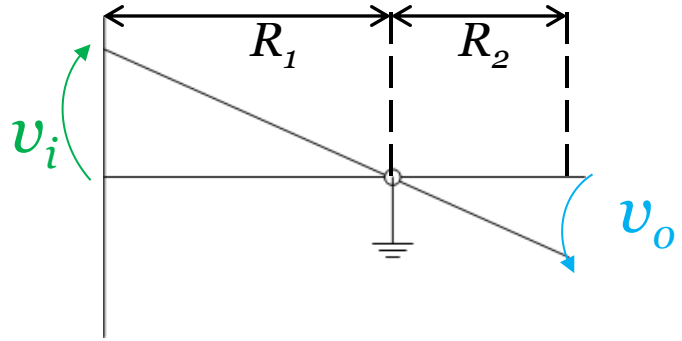
Use the limiting case that $a \to \infty$

The closed-loop gain,

$$A = \frac{v_O}{v_i} = -a\frac{\frac{R_2}{R_1}\beta}{1+a\beta} = -\frac{R_2}{R_1}$$



The mechanical analogy of the inverting amplifier,

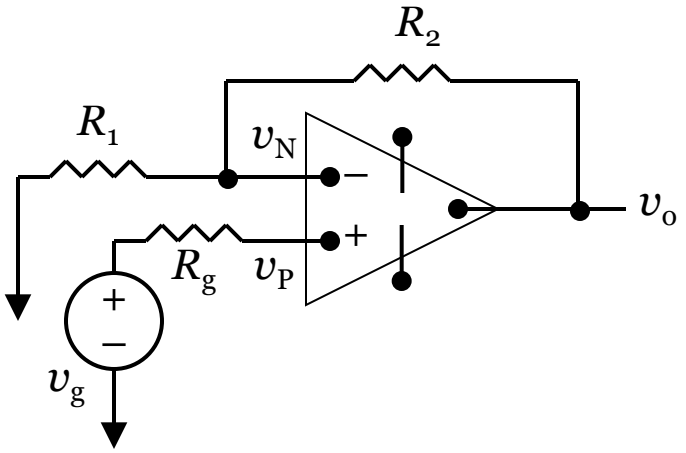


Op Amp Configurations/Non-inverting Amplifier

Analyze using an op amp model with

$r_i = \infty;\ r_o = 0;$ and

gain a = *finite*



Assume it is ideal, apply voltage division on v_N

$$v_g = v_N = v_P = \frac{R_1}{(R_1 + R_2)} v_O$$

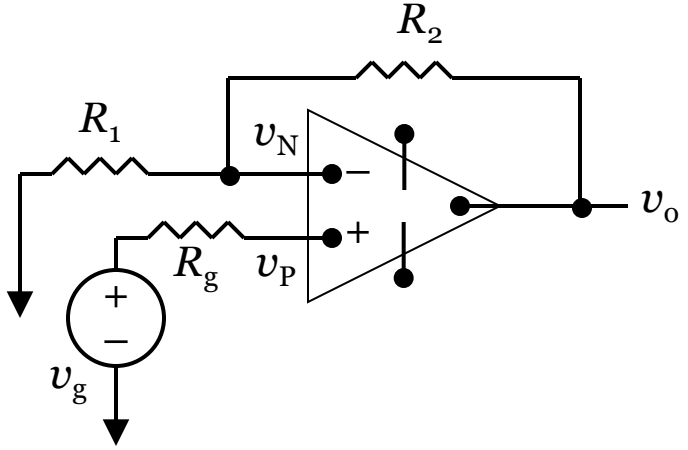
$$v_N = \frac{1}{1 + \frac{R_2}{R_1}} v_o$$

Op Amp Configurations/Non-inverting Amplifier

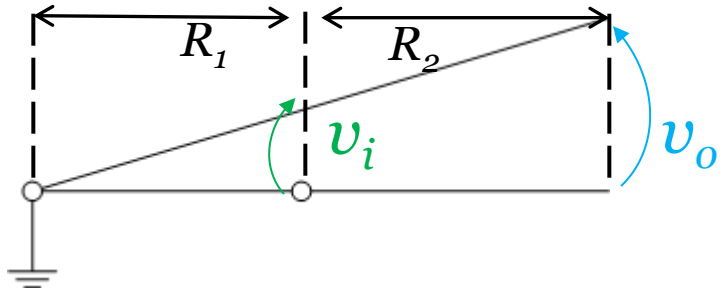
$$v_N = \frac{1}{1+\frac{R_2}{R_1}} v_o \qquad v_o = a(v_P - v_N)$$

$$v_o = a(v_g - v_N) \qquad v_o = a\left(v_g - \frac{1}{1+\frac{R_2}{R_1}} v_o\right)$$

$$A = \frac{v_o}{v_g} = (1+\frac{R_2}{R_1}) \frac{1}{1+(1+\frac{R_2}{R_1})/a}$$



The mechanical analogy of the non-inverting amplifier,



$$A = \frac{v_o}{v_g} = \left(1+\frac{R_2}{R_1}\right) \quad when\ a\ is\ inf$$

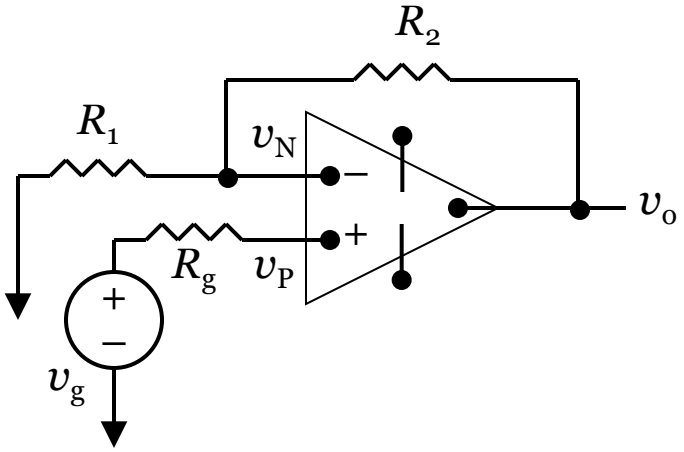
Op Amp Configurations/Non-Inverting Amplifier

Example: An non-inverting op amp with vg = 1 V, and R_1 = 2 kΩ, and R_2 = 18 kΩ. Find

(a) v_o if a = 10^2 V/V

(b) a = 10^4 V/V

(c) a = 10^6 V/V



$$A = \frac{v_o}{v_g} = (1 + \frac{R_2}{R_1}) \frac{1}{1 + (1 + \frac{R_2}{R_1})/a}$$

Op Amp Configurations/Non-inverting Amplifier

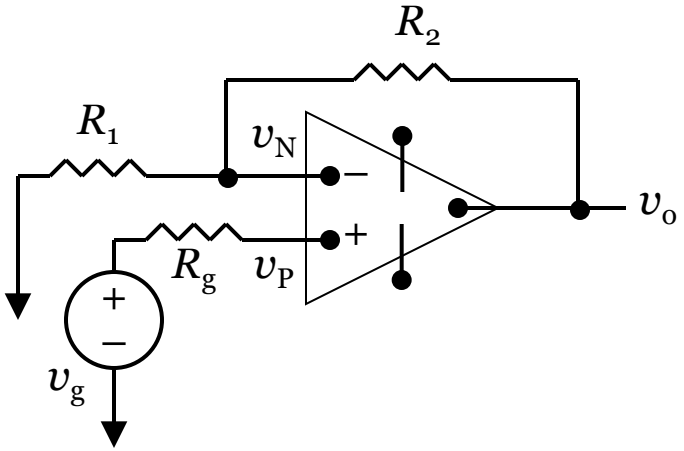
Example: An non-inverting op amp with vg = 1 V, and R_1 = 2 kΩ, and R_2 = 18 kΩ. Find

(a) v_o if a = 10^2 V/V

(b) a = 10^4 V/V

(c) a = 10^6 V/V

Solution



From: $$A = \frac{v_o}{v_g} = (1 + \frac{R_2}{R_1}) \frac{1}{1 + (1 + \frac{R_2}{R_1})/a}$$

(a) $$v_o = 10 \times \left(\frac{1}{1 + 10/10^2} \right) = 9.091V$$

(b) $$v_o = 10 \times \left(\frac{1}{1 + 10/10^4} \right) = 9.990V$$

(c) $$v_o = 10 \times \left(\frac{1}{1 + 10/10^6} \right) = 9.999V$$